

M^5 Day 4: Exponents

Name: _____

1. State whether these statements are True or False. If False, give one example showing why. All variables represent real numbers.

a. $\sqrt{x^2 + y^2} = x + y$.

b. $\sqrt{x^2} = x$.

c. $(x^{-1})^{-1} = x$ (assuming $x \neq 0$).

d. $(x^3)^{1/3} = x$.

e. $\frac{a}{x+y} = \frac{a}{x} + \frac{a}{y}$ (assuming $x \neq -y, x \neq 0, y \neq 0$)

f. $\frac{a+b}{x} = \frac{a}{x} + \frac{b}{x}$ (assuming $x \neq 0$)

2. The following are not True always, but are True sometimes. Find all possible values of the real variables for which these are True:

a. $(a + b)^2 = a^2 + b^2$

b. $\sqrt{a + b} = \sqrt{a} + \sqrt{b}$

c. $(a + b)^{-1} = a^{-1} + b^{-1}$

d. $\frac{x}{1 + x} = x + 1$

e. $\frac{1 + x}{x} = 1$